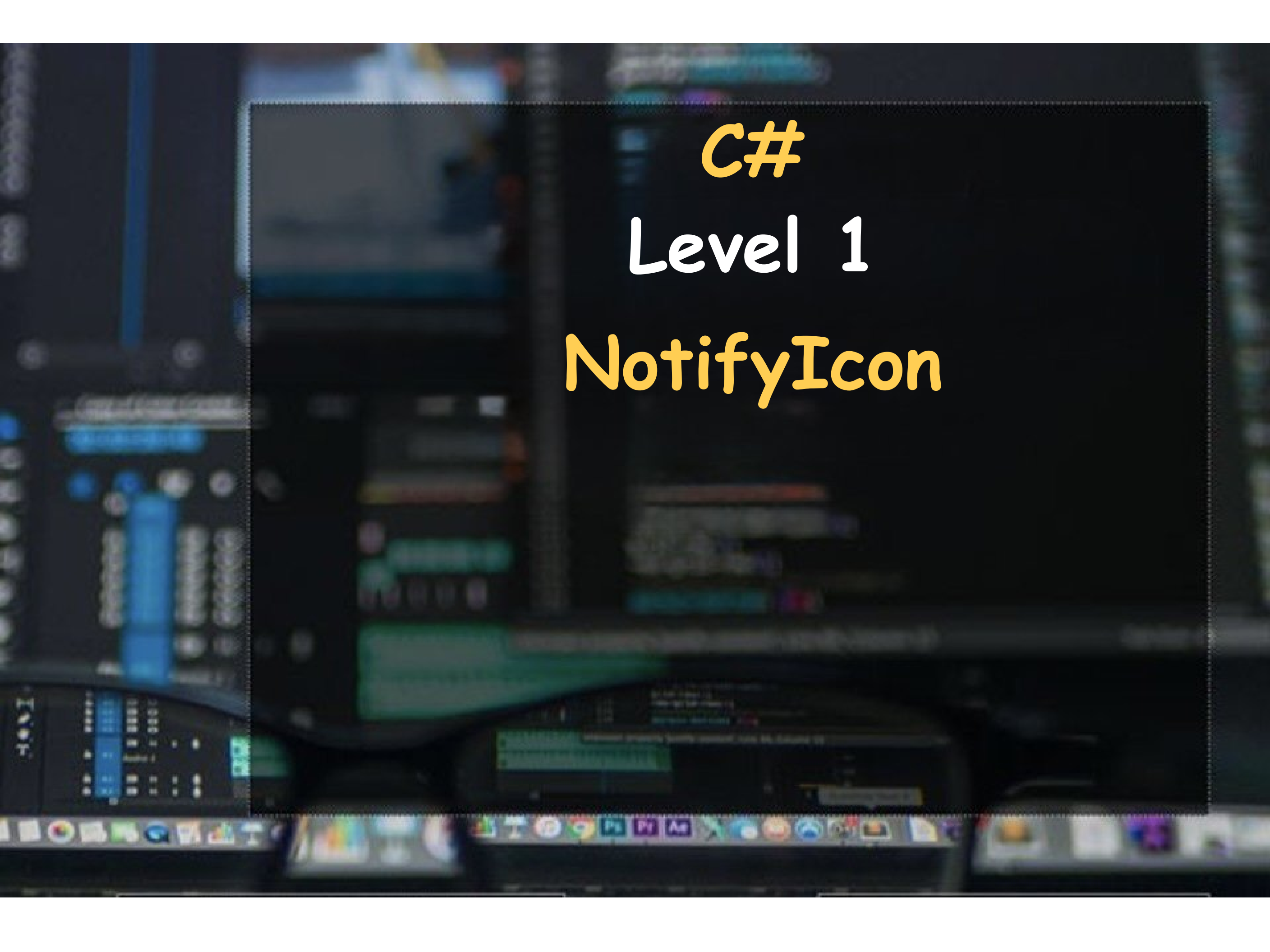




C#

Level 1

NotifyIcon



C#

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NotifyIcon

Definition

NotifyIcon Component

 Windows Forms component that displays an application icon in the Windows notification area (system tray).

 Can display balloon notifications and respond to user interactions.

Explanation

Icon Configuration

-  ``Icon`` specifies the icon displayed in the system tray.

-  ``SystemIcons.Application`` provides a standard Windows application icon.

Balloon Notification

-  ``BalloonTipIcon`` specifies the notification icon type.

-  ``BalloonTipTitle`` sets the notification title.

-  ``BalloonTipText`` sets the notification message.

-  ``ShowBalloonTip()`` displays the notification.

Notification Interaction

-  ``BalloonTipClicked`` detects when the user clicks the balloon notification.

-  Other mouse events such as ``Click`` and ``DoubleClick`` can respond to interaction with the tray icon.

Procedure

1. Add a NotifyIcon component to the Windows Form.
2. Assign an icon using the `Icon` property.
3. Configure the notification icon, title, and message.
4. Call `ShowBalloonTip()` to display the notification.
5. Handle `BalloonTipClicked` if a response to notification clicks is required.

💡 Example

```
private void button1_Click(object sender, EventArgs e)
{
    notifyIcon1.Icon = SystemIcons.Application;
    notifyIcon1.BalloonTipIcon = ToolTipIcon.Info;
    notifyIcon1.BalloonTipTitle = "This is the title.";
    notifyIcon1.BalloonTipText = "This is the message.";
    notifyIcon1.ShowBalloonTip(5000);
}

private void notifyIcon1_BalloonTipClicked(object sender, EventArgs e)
{
    MessageBox.Show("You clicked the notification icon!");
}
```

➡ Result

- ▣ Clicking Button 1 displays a notification in the Windows notification area.
- ▣ The notification contains a title and message.
- ▣ `ShowBalloonTip(5000)` requests a display duration of approximately 5 seconds.
- ▣ Clicking the notification triggers `BalloonTipClicked`.

Note

◆ Important properties:

- `Icon` → system-tray icon.
- `Visible` → controls tray-icon visibility.
- `BalloonTipIcon` → notification icon type.
- `BalloonTipTitle` → notification title.
- `BalloonTipText` → notification message.
- `Text` → tooltip shown when hovering over the tray icon.
- `ContextMenuStrip` → adds a context menu to the tray icon.

◆ `ToolTipIcon` types:

- `Info` → information.
- `Warning` → warning.
- `Error` → error.
- `None` → no notification icon.



Comparison

NotifyIcon

- ✓ Displays an icon in the system tray.
- ✓ Can show balloon notifications.
- ✓ Supports mouse and notification events.
- ✓ Can use a `ContextMenuStrip`.

MessageBox

- ✓ Displays a dialog box inside the application.
- ✓ Supports buttons and message icons.
- ✓ Requires user interaction in many cases.
- ✓ Is not a system-tray notification.

⊘ Warning

- ◆ `NotifyIcon` requires an icon assigned to `Icon` to display correctly.
- ◆ `ShowBalloonTip()` uses milliseconds, e.g. `5000` = approximately 5 seconds.
- ◆ The actual notification duration can be affected by Windows settings.
- ◆ `NotifyIcon` should be disposed when it is no longer needed.

✦ Summary

- ✓ `NotifyIcon` allows a Windows Forms application to interact with the system tray.
- ✓ `Icon` → sets the tray icon.
- ✓ `Visible` → controls icon visibility.
- ✓ `BalloonTipTitle/Text` → define the notification content.
- ✓ `ShowBalloonTip()` → displays the notification
- ✓ `BalloonTipClicked` → handles notification clicks.
- ✓ `ContextMenuStrip` → adds a context menu.

Thank you 😊